

OMKARESHWARI MASHETTE

Data Science Engineer | B.E-Data Science (2026)
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CAREER OBJECTIVE

To build and deploy data-driven and GenAI solutions that improve business outcomes using robust ML practices, clear evaluation, and responsible AI guardrails, while continuously learning modern agentic workflows and MLOps on Azure.

TECHNICAL SKILLS

Programming	Python, SQL, Git
ML	Scikit-learn, XGBoost, LightGBM, Model Evaluation (AUC/F1), Imbalance Handling
GenAI / RAG	RAG, Embeddings, Hybrid Retrieval, Chunking Strategies, Citations, Guardrails
NLP / LLMs	Token limits, context windows, prompt patterns, structured outputs (JSON)
Agentic AI	LangGraph, Tool/Function Calling, Planning & Routing, ReAct Orchestration
Frameworks	LangChain, LlamaIndex, FastAPI, Flask, Pydantic
Vector DB	Azure AI Search (Vector), FAISS, Chroma
Azure	Azure OpenAI, Azure ML, Blob/ADLS, Key Vault, App Service
MLOps / Evals	MLflow, Prompt/LLM Evals, Offline Test Sets, Monitoring Basics
Data	Pandas, NumPy, EDA, Feature Engineering, Data Quality Checks

INTERNSHIP EXPERIENCE

GenAI Intern (RAG-based Systems) — Target Corporate Services Pvt Ltd | 6 Months

- Built a GenAI Scientific Literature Assistant using RAG (LangChain, LlamaIndex, Azure OpenAI).
- Implemented document processing, indexing, and vector-based retrieval pipelines.
- Applied prompt engineering for accurate, citation-based and grounded responses.
- Improved response quality using LLM evaluation techniques and offline test sets.

KEY RESPONSIBILITIES

- RAG & Retrieval:** Designed GenAI solutions using RAG patterns with grounded responses, citations, and fallback behavior; improved faithfulness via chunking, metadata filters, and evaluation sets.
- Agentic Workflows:** Built agentic workflows with tool/function calling, routing, and verification steps; defined tool schemas and structured outputs to reduce hallucination risk.
- Prompt Engineering:** Engineered prompts for consistent JSON outputs and safe responses; applied prompt versioning and offline tests to measure quality, latency, and cost.
- ML Modelling:** Developed classical ML classification models with robust validation, leakage checks, threshold optimization, and SHAP-based explainability.
- MLOps:** Implemented MLflow experiment tracking, reusable training pipelines, and batch scoring; documented versioning decisions for reproducibility.

PROJECTS

1. GenAI Scientific Literature Assistant (RAG)

Client: NovaPharm, US |

Industry: Pharma |

Role: GenAI Engineer

Tools: Python, Azure OpenAI, LlamaIndex, LangChain, Azure AI Search (Vector), Prompt Engineering, Evaluation
Developed a GenAI assistant for scientific literature summarization and Q&A, grounded on publications and internal documents with citations and claim verification steps.

- Built ingestion and indexing pipelines for PDFs/abstracts with chunking tuned for scientific sections (methods/results).
- Implemented citation-first prompts and constrained output formats to reduce unsupported claims.
- Added query decomposition and multi-hop retrieval for complex questions spanning multiple sources.
- Created evaluation harness with factuality checks and human review workflow for high-impact summaries.
- Enabled secure configuration, secret handling, and documented governance for prompt and index updates.

2. ShadowLock– AI-Enhanced Steganography Platform

Client: Academic Project |

Industry: Cybersecurity / Data security |

Role: AI/ML Engineer

Tools: Python, NumPy, OpenCV, Scikit-learn, Flask / Django, MySQL / MongoDB, Jupyter Notebook / VS Code

Concepts & Techniques: Steganography, Machine Learning, Image Processing, Encryption

Developed an AI-enhanced steganography platform to securely hide and transmit sensitive data within digital images. The system integrates encryption with machine learning techniques to improve data security, reduce detectability, and ensure safe communication.

- Built a secure web application to hide sensitive data inside images using LSB steganography.
- Integrated AI-based mood detection and encrypted private journaling features.
- Developed backend APIs using Flask to handle secure image upload, encoding, and decoding processes.

EDUCATION

B.E — Computer Science & Engineering (Data Science)

BKIT Engineering College (VTU University)

Expected 2026

CERTIFICATIONS

- Programming in Python — Echo Brains
- NPTEL Certification — Introduction to Artificial Intelligence

DECLARATION

I hereby declare that all the above particulars and information furnished are true to the best of my knowledge.

Place: _____

Date: _____

Omkareshwari Mashette